Fakturama Image-to-Cash Automation

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1. Scope and success criteria

Turn one order image into one saved Order and one linked Invoice in Fakturama 2.2.0 on Windows, through its user interface. Success means the persisted records were read back and match the source, not that the clicks succeeded. The Order must remain open; the Invoice must carry the source payment method and, only for a PAID order, its payment date and full value. No Delivery, Correction or Dunning document is ever created.

Supported scope: EUR; Bank Transfer, Credit Card or SEPA Direct Debit; PAID with a date or UNPAID without one; zero order-level discount and shipping. Anything else (another currency, a partial payment, a nonzero overall charge, unreadable fields) stops for review. The design is a verified vertical slice first, then the missing-master branches, then multi-line orders and recovery.

2. Architecture and extraction

Four layers, so business rules can be tested without Fakturama:

- **Extraction.** Local, bounded PNG/JPEG input (size, dimensions and frame checks, orientation applied, re-encoded to strip metadata). Tesseract OCR gives local text; an optional vision model (Claude, or a local Ollama model) returns data constrained by a JSON schema. Remote use is opt-in and sends only the order image. The supplied image is 385x529 px, so OCR reads the header, payment row and printed totals but not the item table; OCR is therefore used as an independent cross-check of a candidate extraction, not as the source of the item lines.
- **Validation.** Strict immutable schema (unknown fields, control characters, non-finite or negative amounts and impossible dates are rejected) and decimal finance: line net is quantity x unit net x (1 - discount%); VAT is aggregated by rate and rounded per rate; the Product master gross price is unit net x (1 + VAT%), rounded to cents, and never includes the line discount. Totals must reconcile with the image before anything is typed. A reconciled total does not prove a correct name or SKU, so a human confirms the extraction.
- **Workflow.** Order-first sequence with exact, unique matching (Company, First name, Name, ZIP, City for a Debtor; SKU for a Product). Missing records are created only after an exact selector check returns nothing.
- **Windows adapter.** Python with pywinauto (UI Automation), plus Win32 calls for window identity and the clipboard.

Image text is data, never instructions: the extractor has no tools and its output is parsed against the schema before use.

3. Desktop grounding and the continuous workflow

The adapter binds to the Fakturama process found by its executable name, requires it in the foreground, and re-checks identity before each action. Controls are found by their UIA type, label and position in the tree, scoped to the current editor or dialog, and must match exactly once. Fakturama's SWT item and result grids expose no UIA rows or cells, so the adapter reads them with Select All / Copy from a clipboard freshly owned by Fakturama (two identical copies), and locates grid headers with OCR on a fresh capture of the control. No screen coordinate is stored. UIA patterns (value, invoke, toggle) are preferred over keystrokes; where a widget needs them (the segmented date field, a click to give a grid or a popup focus) only fixed navigation keys or clicks computed from live bounds are used.

1. Open New Order first, keep its proposed number, set Date, Cust.Ref., Net and With VAT.
2. In the Order's address selector, search the company and copy the result rows. One exact row is selected; none means the Debtor is missing; several stop the run. An empty grid is recognised from its pixels, because copying from an empty grid makes Fakturama throw.
3. For a missing Debtor, open New Contact while the Order stays open, keep the proposed customer ID, fill the fields, address roles, separate delivery address, alias, Net and the payment method, save once, return to the same Order and re-select the saved Debtor. A missing payment term or VAT record is created first with exact names and values (an operator step in the current implementation).

4. For every line, search the Product selector by exact SKU. Fakturama closes the dialog and adds the line itself when exactly one product matches. A missing Product is created with the source SKU and description and the computed gross price, saved once, and re-selected from the same Order. Quantity, unit net and discount are then set and read back per line.
5. Verify both address pages, all lines and the totals. The Invoice address tab must be selected before saving: Fakturama 2.2.0 throws while saving an Order with the Delivery tab selected. Save once and confirm the Order in Data > Documents (open state, date, reference, total).
6. From the saved Order's follow-up area create the Invoice (never the toolbar). Verify the copied Order Date, both addresses, every item and the totals, apply the payment method, `paid`, the payment date and the Value for a PAID order, save once, and confirm in Data > Documents that the Invoice is paid and the Order is still open.

4. Recovery, security and validation

Every live command writes an intent file before acting and a report afterwards. Save, Invoke and creation actions run once and are never retried: only observations are retried. A stop raises a coded review error with the progress flags, and recovery from an uncertain result is manual; accounting records are never deleted or rolled back automatically. The simulator workflow additionally uses a SQLite journal with a hashed business key and a single-writer lease to block duplicate or altered replays.

Credentials come from the environment. Only the opt-in extraction call leaves the machine. Fakturama's database is never written to; its log is read only to cross-check persistence. See docs/security.md.

Unit tests cover decimal rounding, multi-rate and multi-line totals, invalid extraction, matching conflicts and the payment mappings (Bank Transfer to Credit transfer, Credit Card to Credit card, SEPA Direct Debit to SEPA direct debit). Recorded UIA fixtures cover the Windows bindings, and a simulator runs success and failure scenarios (missing, existing, mixed and duplicate masters, a failure after the Order Save, a corrupted total). Simulator and fixture results do not prove compatibility with Fakturama; that is shown by the live runs recorded in docs/live-run.md.